

◆ Gemini

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import cv2
import matplotlib.pyplot as plt

# NOTE: You need to upload your image file (e.g., 'image1.jpg') to the Colab environment fir
# For example, by dragging and dropping it into the files section on the left sidebar.
image_path = 'image1.jpg' # Replace with the actual filename of your uploaded image
image_path = 'wp3892792.jpg' # Replaced with the actual filename of your uploaded image
img = cv2.imread(image_path)

# Check if the image was loaded successfully
if img is None:
    print(f"Error: Could not load image from {image_path}. Please ensure the file exists and
else:
    gray_img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Display the grayscale image (using matplotlib for better Colab integration)
    plt.imshow(gray_img, cmap='gray')
    plt.title('Grayscale Image')
    plt.axis('off')
    plt.show()

    cv2.imwrite('gray_image.jpg', gray_img)

# cv2.imshow and cv2.waitKey are typically used for local desktop applications,
# and might not work as expected or cause issues in Colab's web environment.
# For displaying images in Colab, matplotlib.pyplot is generally preferred.
# If you still want to use cv2.imshow (e.g., for debugging in a local environment),
# ensure your environment supports it (e.g., not a headless server).
# For Colab, commenting them out or replacing with matplotlib is recommended.
# cv2.waitKey(0)
# cv2.destroyAllWindows()
```

Grayscale Image

